

JS / KL Divergence Analysis

Epoch-wise Divergence from Baseline Distribution

Run Parameters

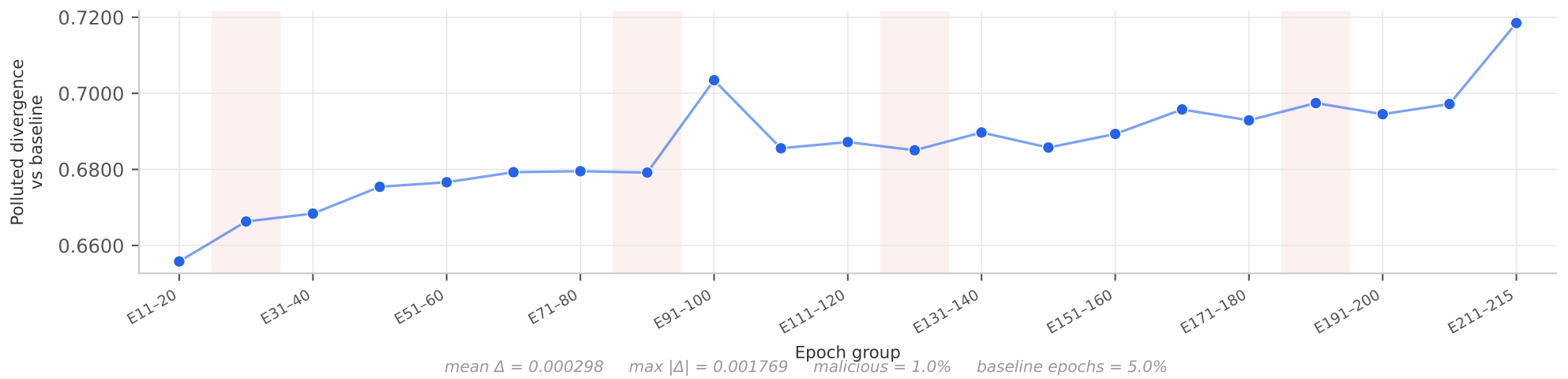
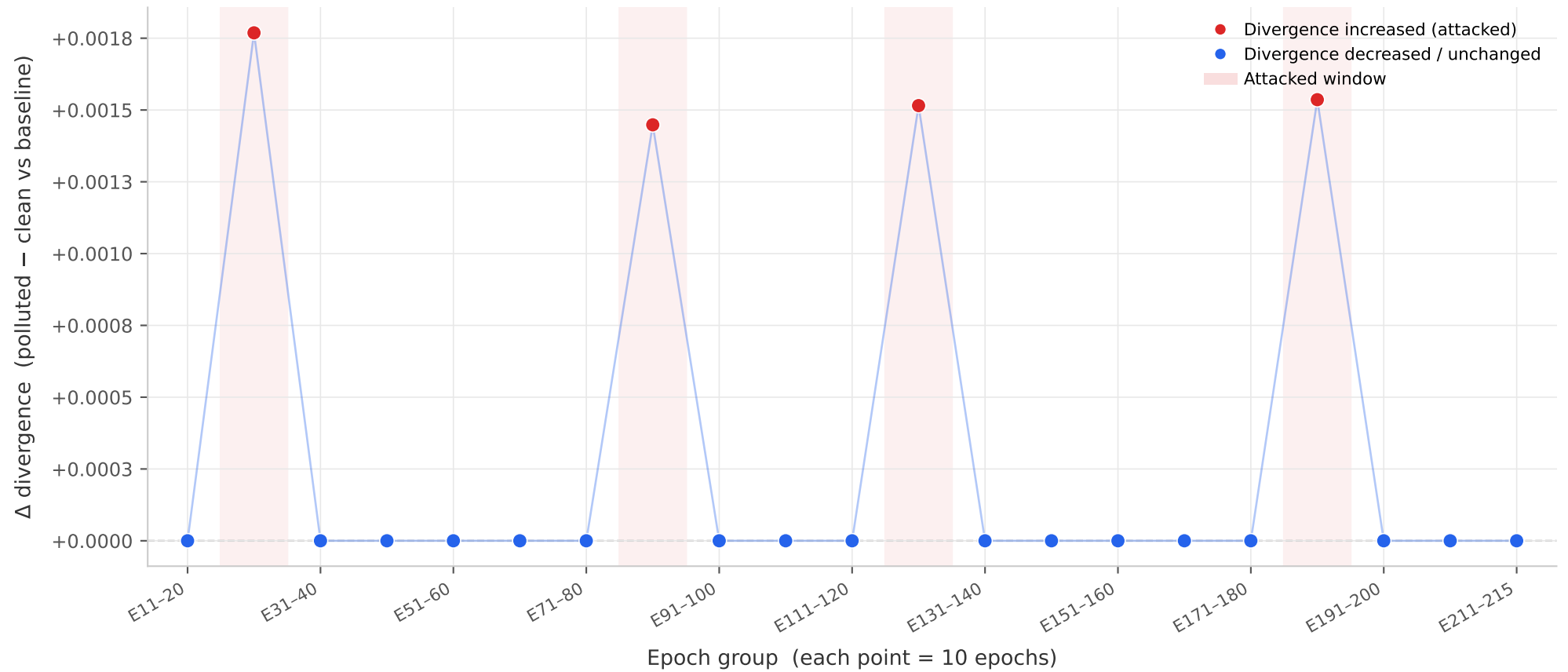
Dataset	epochs_5.0_1.0_sparse.json
Directory	CAIDA_Dataset/txtD280/epochs/
Timestamp	2026-02-19T19:17:03.538192
Method	Jensen-Shannon + KL
Total epochs	215
Baseline epochs	10 (5.0%)
Attack epochs	205
Malicious traffic	1.0%
Epoch group size	10 epochs per point
Metrics plotted	10
Mode	SPARSE
Attack probability	0.3
Windows total	21
Windows attacked	4
Windows clean	17

Baseline Distribution Summary

Total Flows	50558
Unique Src IPs	14859
Unique Dst IPs	10585
Unique Src Ports	9030
Unique Dst Ports	20334
Src IP Vocab Size	14859
Dst IP Vocab Size	10585
Src Port Vocab Size	9030
Dst Port Vocab Size	20334

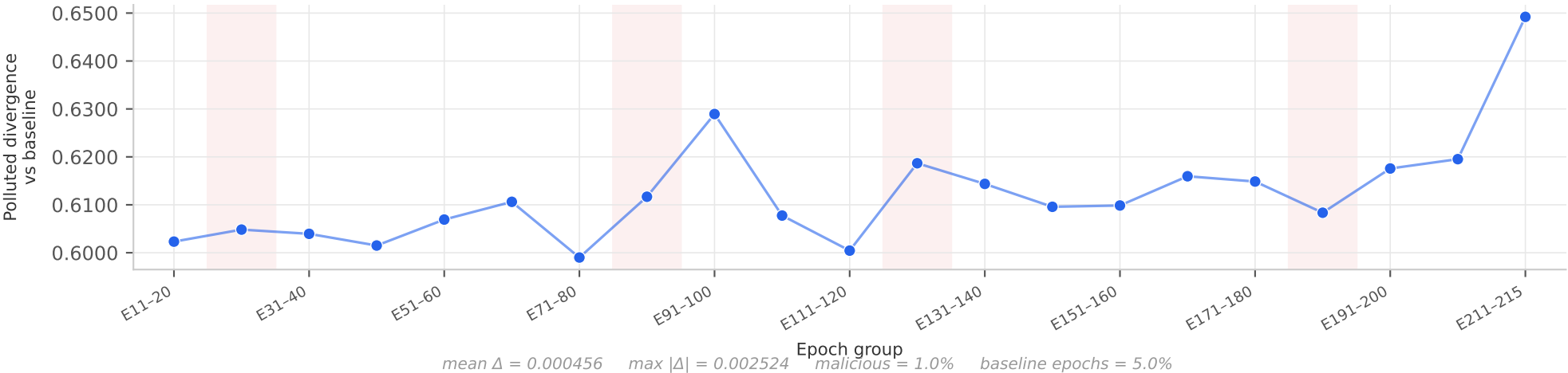
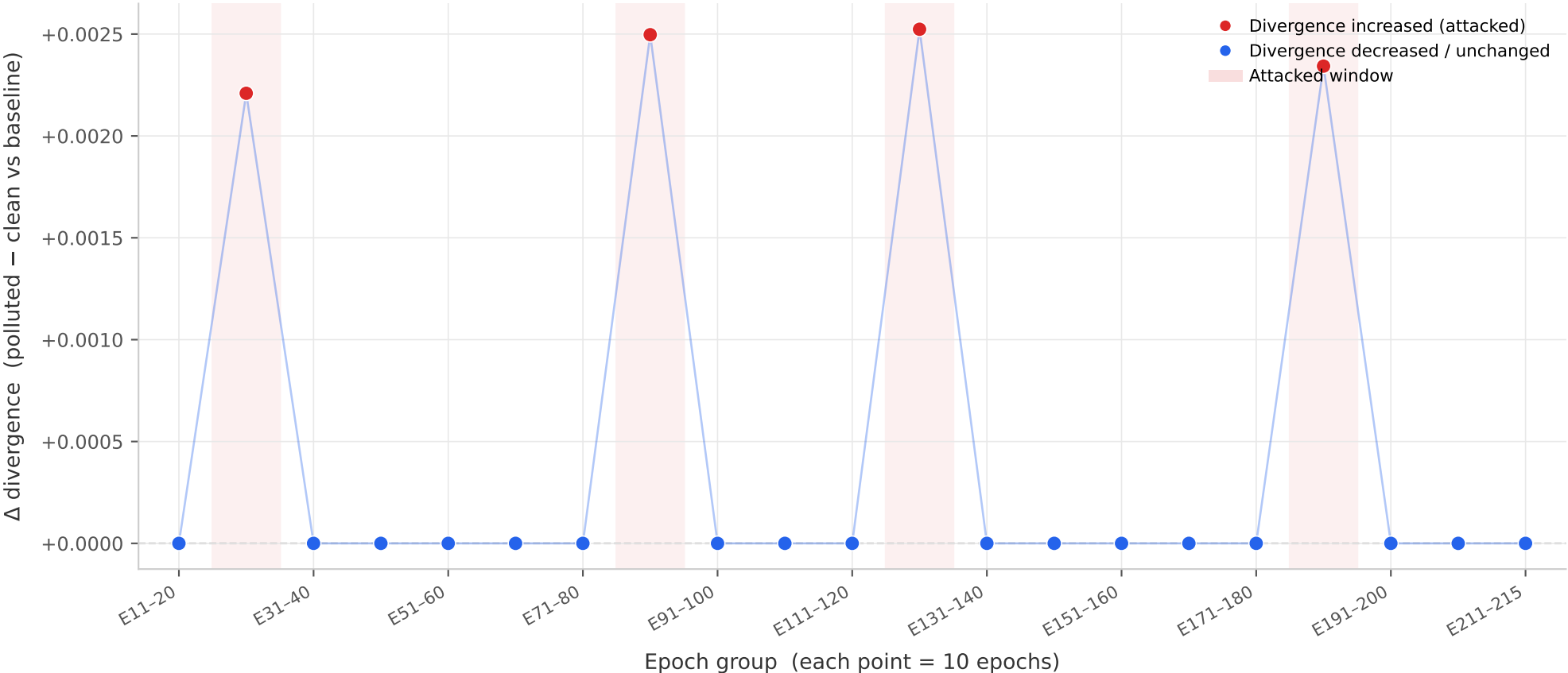
Source IP — JS Divergence

JS Divergence



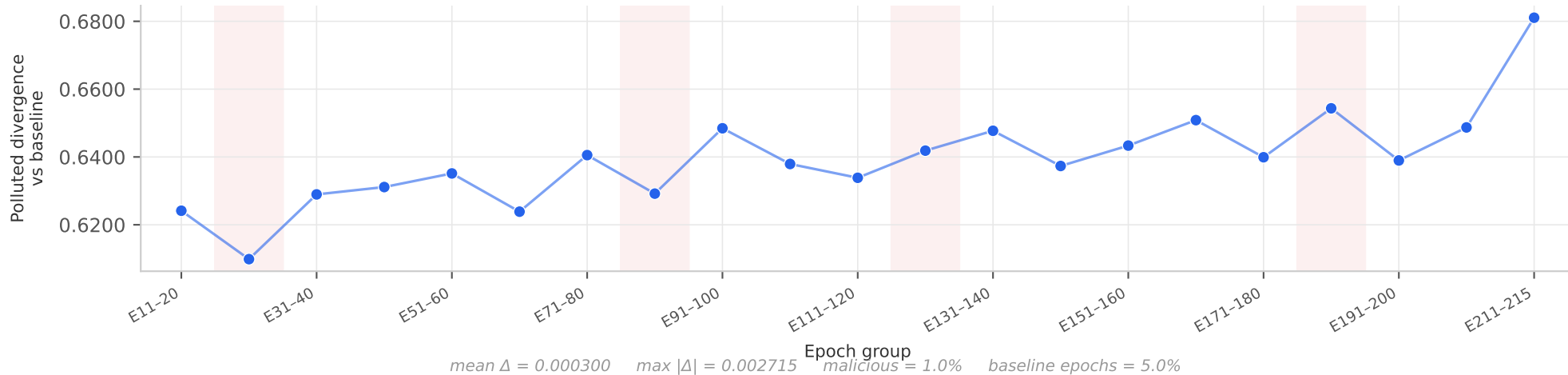
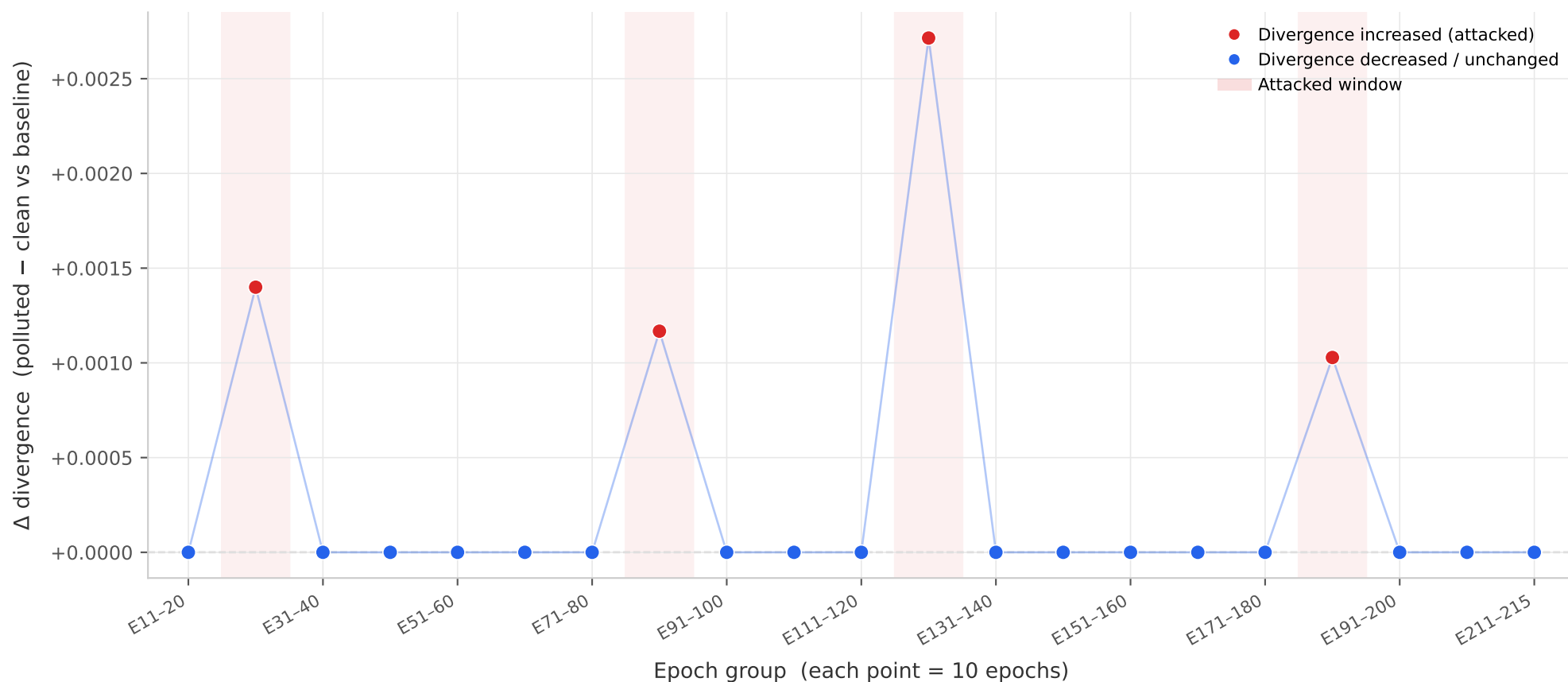
Destination IP — JS Divergence

JS Divergence



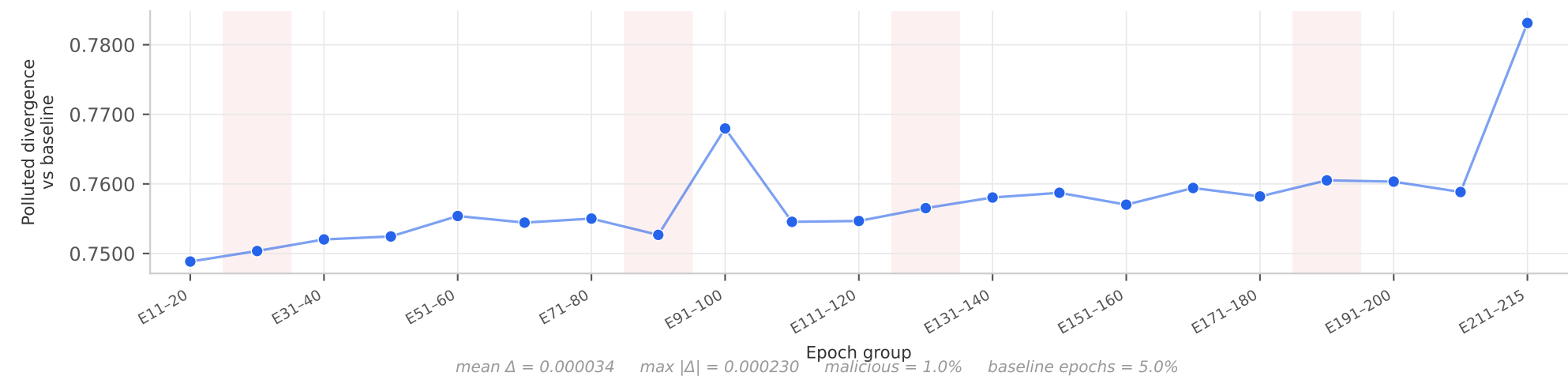
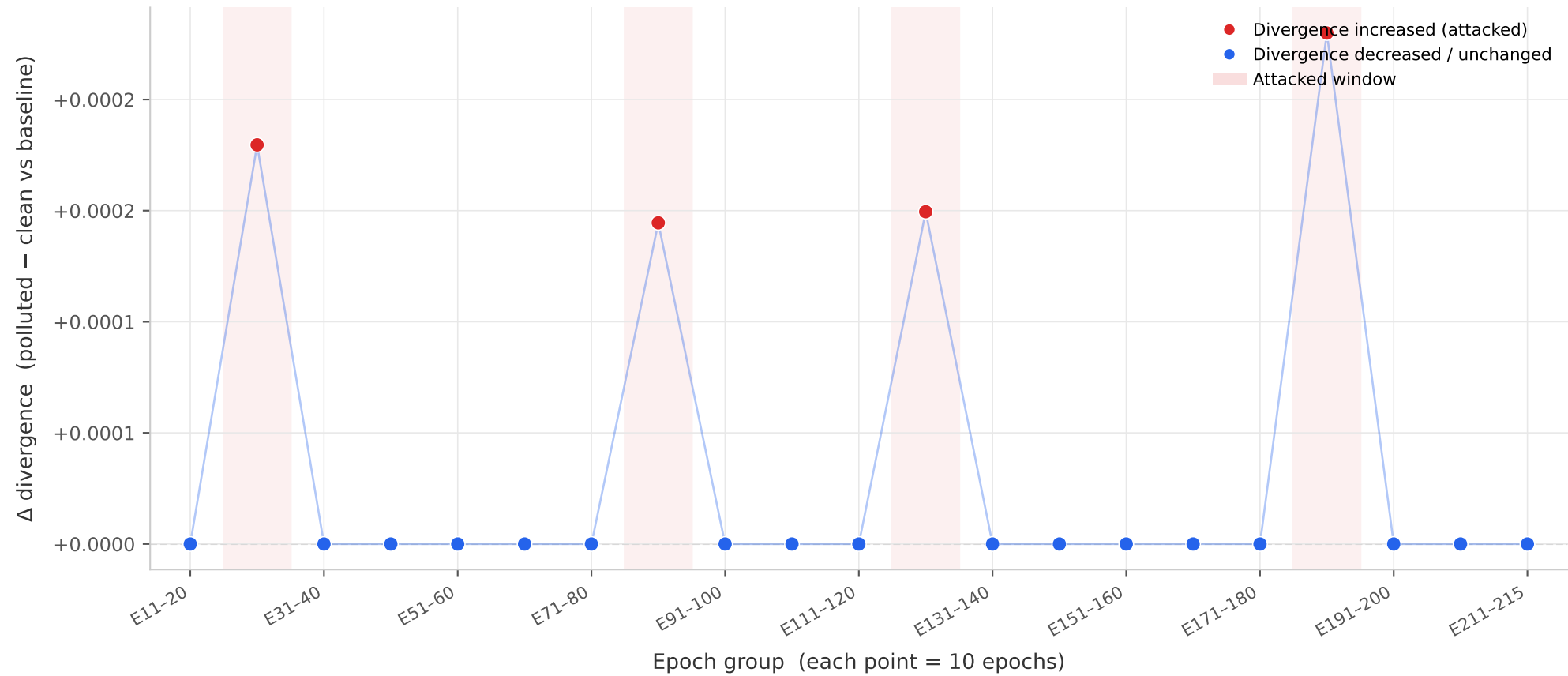
Source Port — JS Divergence

JS Divergence



Destination Port — JS Divergence

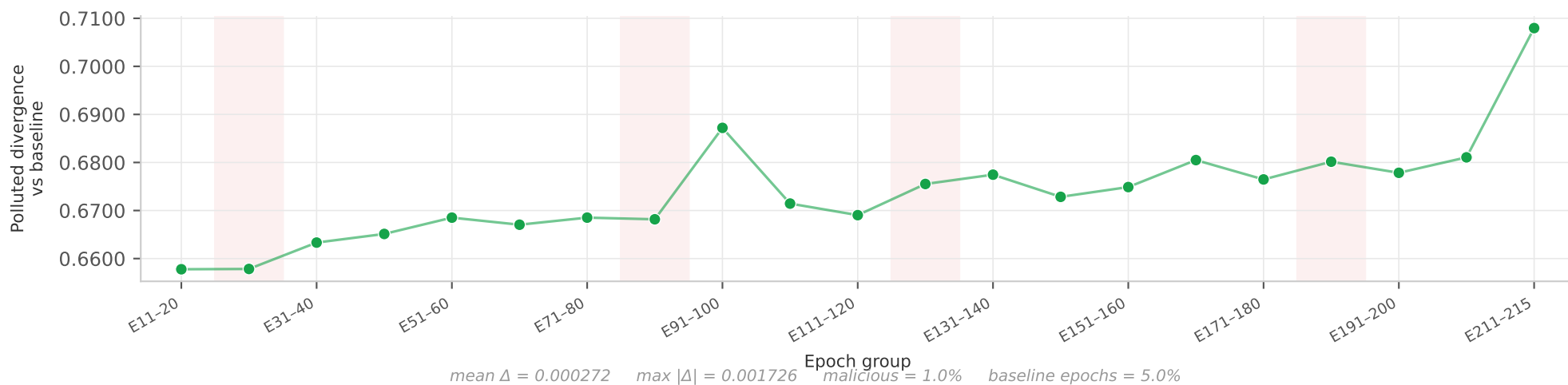
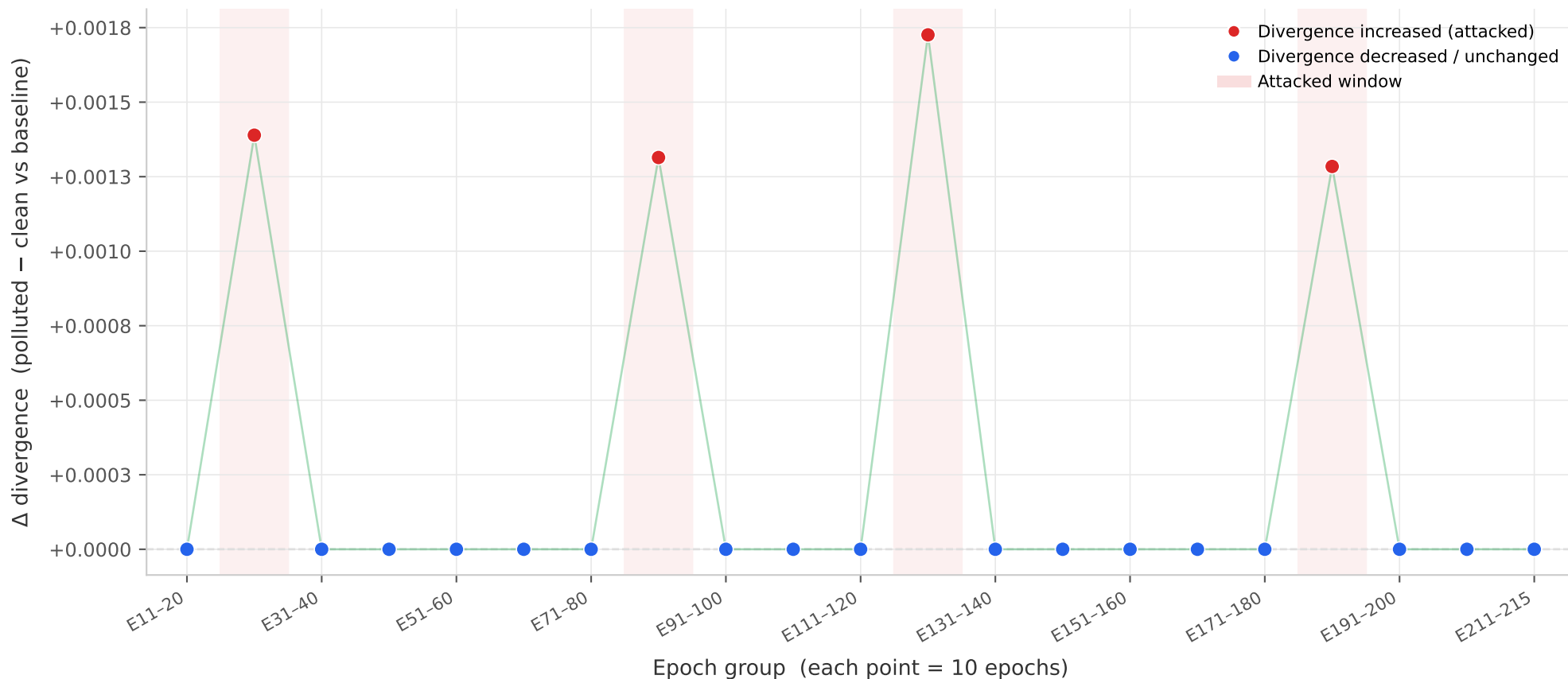
JS Divergence



mean Δ = 0.000034 max $|\Delta|$ = 0.000230 malicious = 1.0% baseline epochs = 5.0%

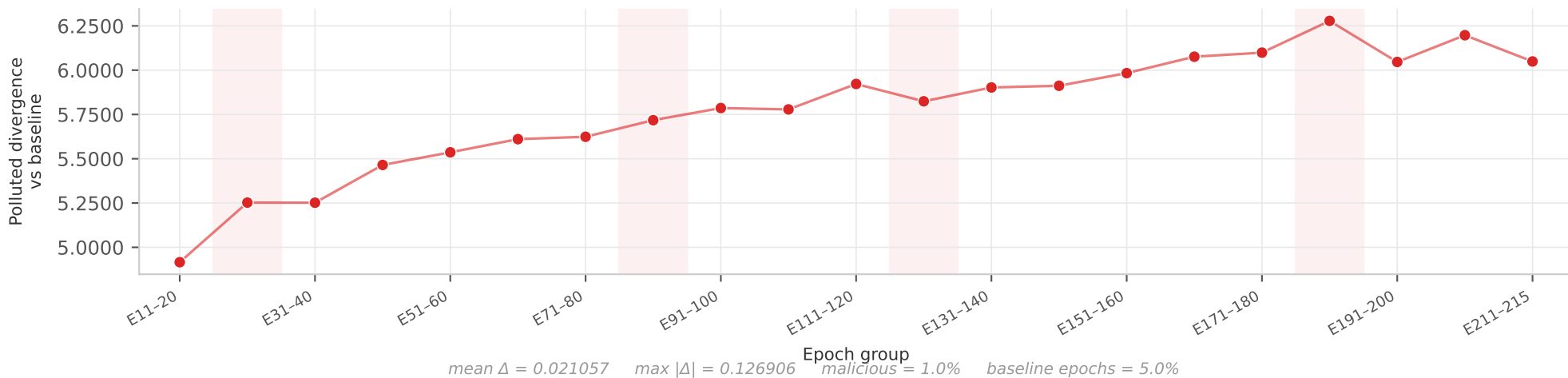
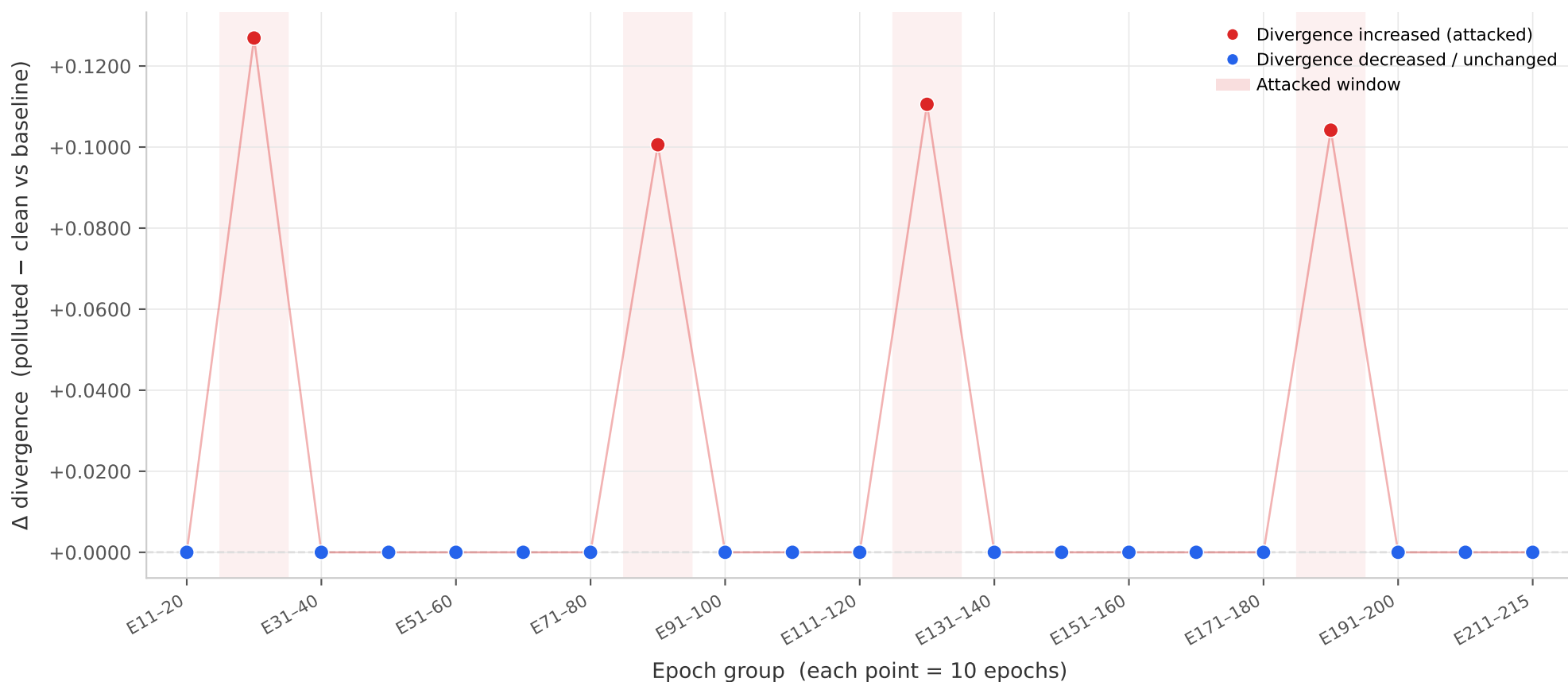
Combined (all features) — JS

Combined



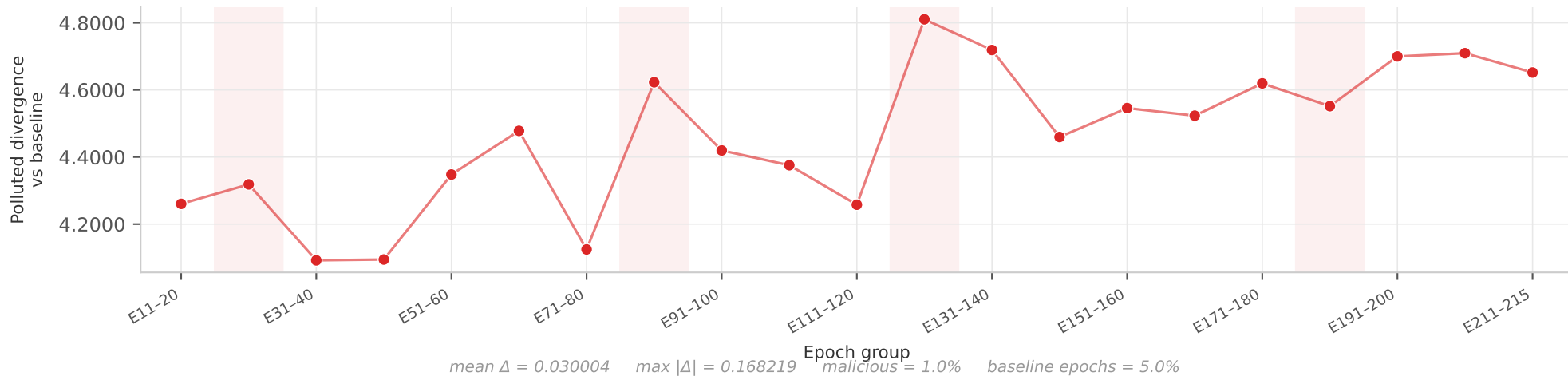
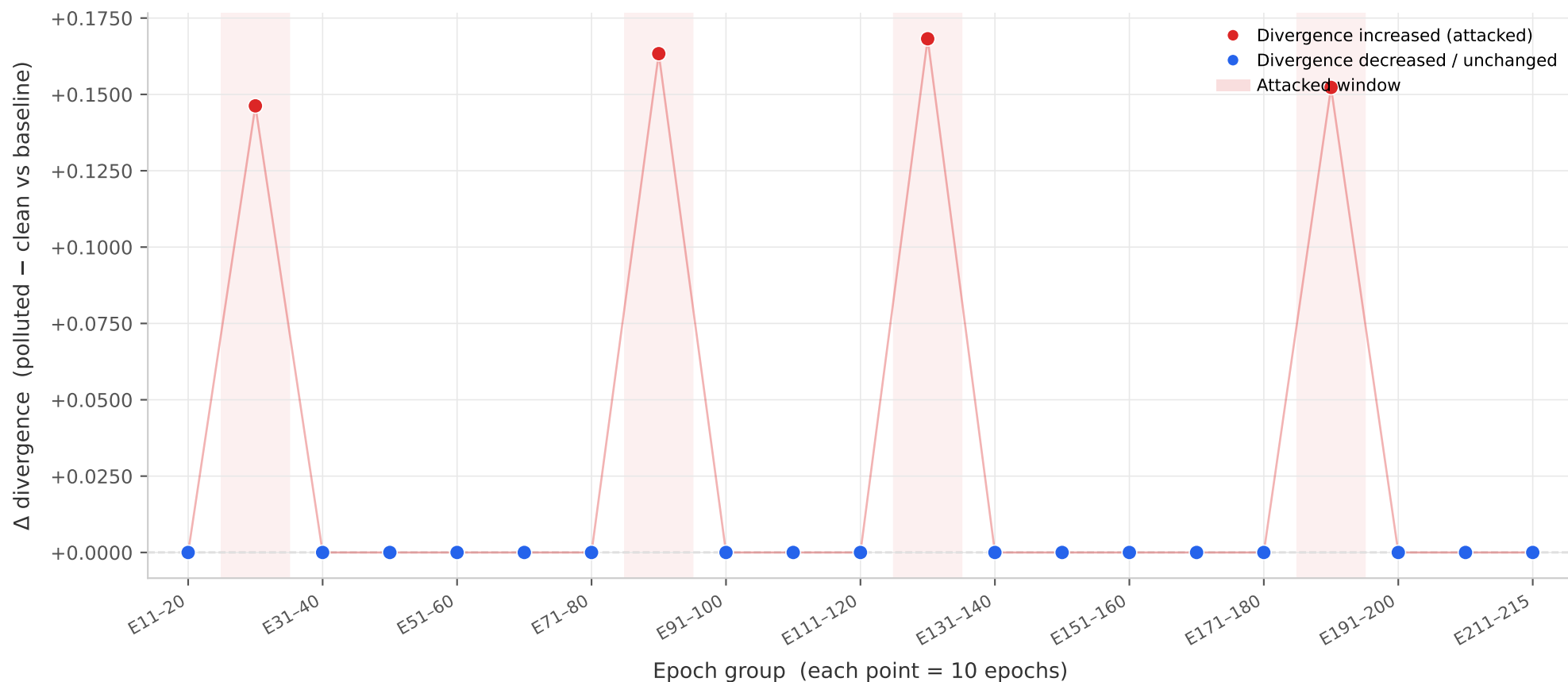
Source IP — KL Divergence

KL Divergence



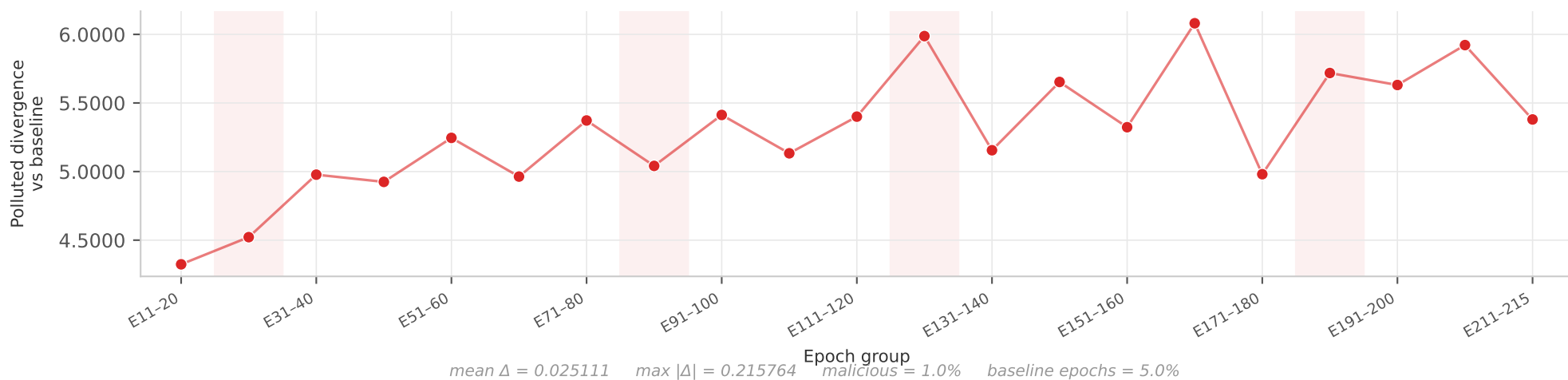
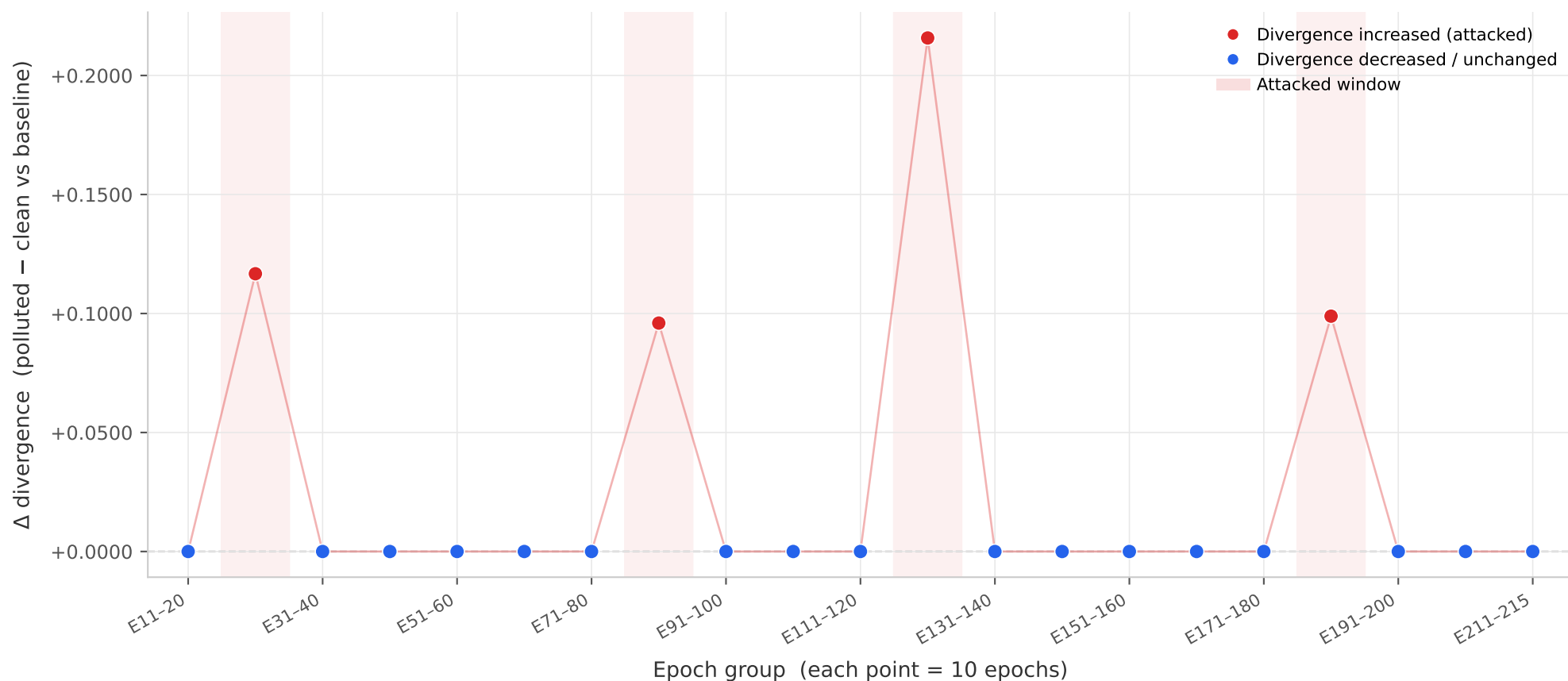
Destination IP — KL Divergence

KL Divergence



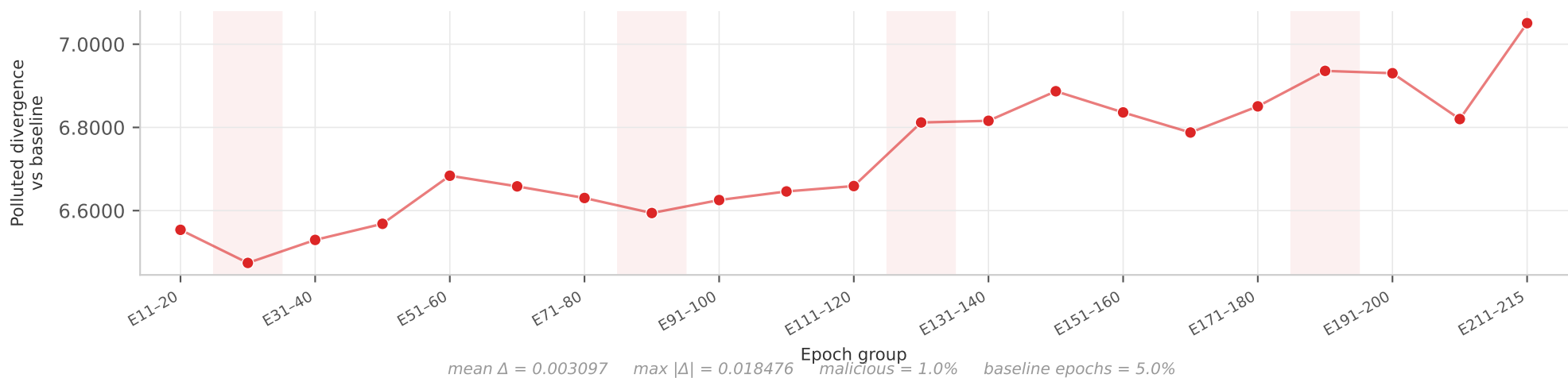
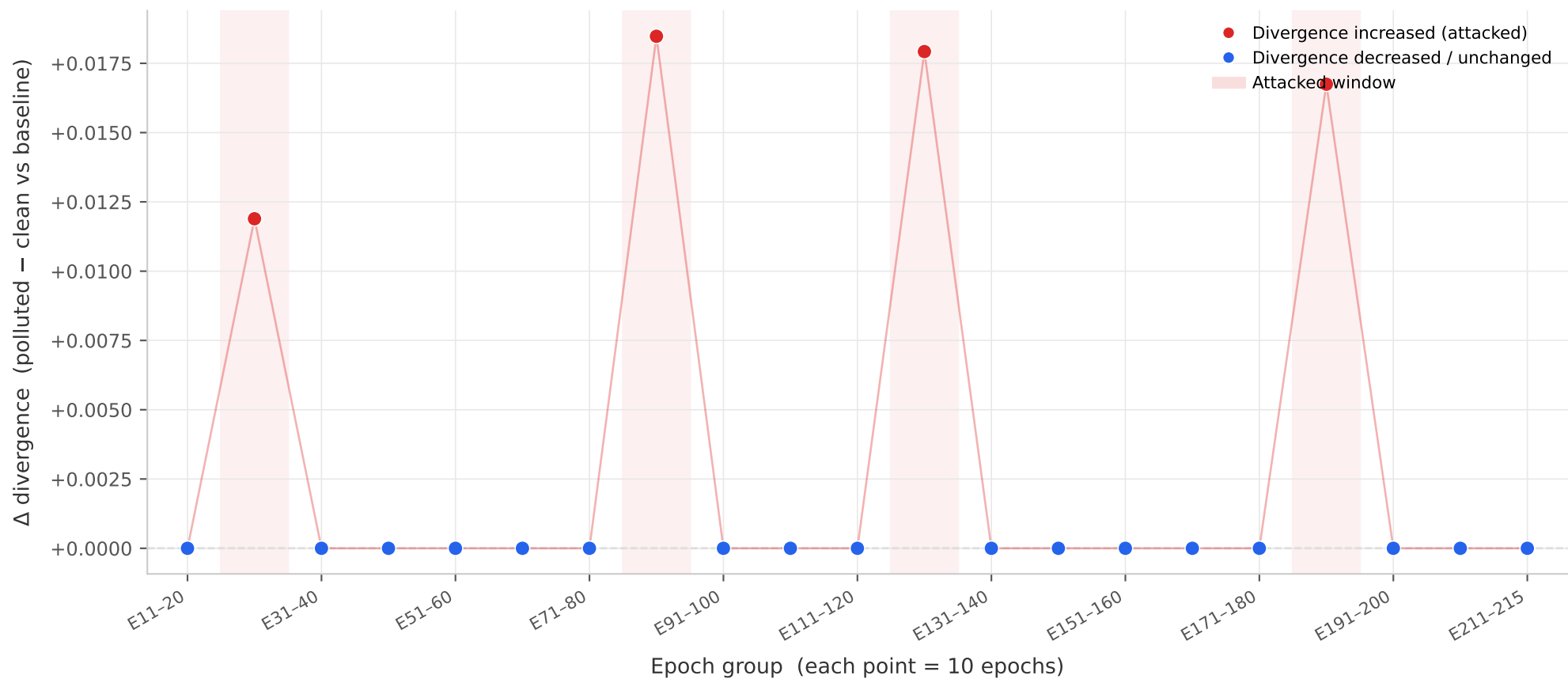
Source Port — KL Divergence

KL Divergence



Destination Port — KL Divergence

KL Divergence



Total Flows

Flow

